

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		Any object [wholly or partially] immersed in a fluid experiences an upwards force equal to the weight of the fluid displaced by the object	1			1		
		(ii)		Sub into $\rho = \frac{m}{V}$ with conversion i.e. 80×10^{-6} and 1 030 (1) Manipulation and answer $(1\,030 \times 80 \times 10^{-6}) = 0.0824 \text{ kg}$ (1)		2		2	2	
		(iii)		Mass of block of ice = 0.0824 kg (can be inferred) (1) Allow ecf from (i) Volume of block of ice = $\frac{0.0824}{920} = 89.6 \times 10^{-6} [\text{m}^3]$ (1) Volume of ice above surface = $9.6 \times 10^{-6} [\text{m}^3]$ (1)	1	1 1		3	2	
		(iv)	I	$\frac{2.2 \times 10^{11} \text{ m}^3 \text{ yr}^{-1}}{3.6 \times 10^{14} \text{ m}^2} [= 6 \times 10^{-4} \text{ m yr}^{-1}]$		1		1		
			II	Melting ice sheets adds to the volume of the sea (1) Melting icebergs replaces volume of sea water already displaced so water level unchanged / change is negligible (1)		2		2		
			III	More radiation absorbed by darker surfaces that is uncovered or less radiation reflected as white reflective surfaces have melted			1	1		
	(b)	(i)		All units correctly identified: Power = W; $[A] = \text{m}^2$; $[\Delta\theta] = \text{K}$; $[\Delta x] = \text{m}$ (1) Convincing algebra e.g. $K = \frac{\Delta Q \Delta x}{\Delta t \Delta \theta A}$ (1)	2			2	1	